

Experiment: Implement Principal Component Analysis

1. Setup Environment

- Open Python Notebook (Jupyter/Colab).
- Import required libraries: scikit-learn, pandas, matplotlib, numpy.

2. Load Dataset

- Select a dataset
- Identify features and target variable.

3. Preprocess Data

- Handle missing values if any.
- Standardize the features (important for PCA).
- Split dataset into training and testing sets.

4. Apply PCA (Principal Component Analysis)

- Choose the number of principal components to retain (e.g., 2 or enough to capture 95% variance).
- Transform the dataset into principal components.
- Visualize data distribution in reduced dimensions.

5. Initialize SVM Classifier

- Select kernel type (linear, RBF, polynomial).
- Set parameters like C and gamma based on problem requirements.

6. Train the Model

- Train the SVM classifier on PCA-transformed training dataset.

7. Make Predictions

- Use the trained model to predict outcomes on PCA-transformed test dataset.

8. Evaluate Model Performance

- Calculate accuracy, precision, recall, and F1-score.
- Generate and analyze confusion matrix.
- Compare results of SVM with and without PCA.

9. Visualization

- Visualize decision boundaries using the reduced PCA features (for 2D).
- Plot comparison charts of model performance with and without PCA.

10. Summarize Results

- Report whether PCA improved performance or computational efficiency.
- Discuss the trade-off between dimensionality reduction and accuracy.